

EECS 498 Project Proposal



Project Idea: Custom Neural Network Library

1 Motivation

Deep learning is a popular machine learning technique due to its ability to efficiently learn patterns from large amounts of data. In an effort to better understand this algorithm, we plan to build our own fully connected neural network library from first principles.

From the theoretical side, this project can guide us on the intricacies of how neural networks work. By taking control of the process of designing each aspect, general phenomena can be further analyzed across each layer of the network, such as the vanishing gradient problem and the “modern regime” (generalization with overparameterization).

On the implementation side of things, we hope that this project will help us understand the procedure for training and testing NNs. This project will also be good practice writing efficient, well-abstracted code with Python (and other scientific tools such as NumPy and PyTorch).

2 Supporting Materials

In addition to the posted lecture slides, there are many online guides^[1] that cover building neural networks from scratch that can be consulted for reference. Online datasets, such as MNIST^[2], can be used to evaluate our network.

3 Plan

We plan to design a fully-connected neural network library from scratch. After developing our own implementation, we hope to compare it to an “out-of-the-box” neural network from a well-known library such as PyTorch. A rough outline of our plan is presented below.

1. Build neural network library from scratch:
 - Design layers
 - Forward-propagation and Back-propagation algorithms
 - Training/testing procedures
2. Compare against black-box model (such as from PyTorch)
3. Attempt to observe phenomena (time permitting) such as:
 - Vanishing gradient (including ways to mitigate)
 - Bias-Variance trade-off

4 References

1. <https://towardsdatascience.com/math-neural-network-from-scratch-in-python-d6da9f29ce65>
2. <http://yann.lecun.com/exdb/mnist/>